```
const ProductSchema = z.object({.
  name: z.string(),
  description: z.string(),
  price: z.number(),
  category: z.string()
});

const ProductListSchema = z.array(ProductSchema);

async function generateProductList(prompt) {
  'use server';
  const {
    object: { products },
  } = await generateObject({.
    model:  google("models/gemini-2.0-flash"),
    schema: z.object({
      products: ProductListSchema.
    }),
    prompt: `Generate a list of 5 products
related to: ${prompt}. Provide name, description, price,
and category for each product.`,
  });
  return products;
}

generateProductList('A list of cereal types').then(console.table);
```

Defines a Zod schema for a single product, specifying the expected structure and types

Creates a schema for an array of products using the ProductSchema

Uses generateObject to create structured data from the AI response, destructuring the result

Applies the ProductListSchema to ensure the AI generates an array of products

Constructs a detailed prompt for the AI, including the user's input and specific instructions

The `generateObject` function handles the communication with the AI model and ensures that the response conforms to our defined schema. This provides us with type-safe, structured data that we can directly use in our application without additional parsing or type-checking.

If you run this example code, it will print something similar in the console (figure 4.12). We printed the results using `console.table`, which renders the data as a table. Since the data is returned in a structured format, we can feed it to other services that expect the same model types.

```
% node example.mjs
```

(index)	name	description	price	category
0	'Honey Nut Cheerios'	'Delicious honey-flavored cereal with a hint of nuttiness'	3.99	'Breakfast Cereal'
1	'Frosted Flakes'	'Crunchy corn flakes coated with a sweet frosting'	2.99	'Breakfast Cereal'
2	'Cinnamon Toast Crunch'	'Toasted squares with a cinnamon and sugar coating'	4.49	'Breakfast Cereal'
3	'Lucky Charms'	'Colorful marshmallow and oat cereal with a touch of magic'	3.79	'Breakfast Cereal'
4	'Raisin Bran'	'Bran flakes with sweet raisins for a nutritious breakfast option'	3.29	'Breakfast Cereal'

Figure 4.12 Structured data generation using the `generateObject` function. The results of the prompt are printed using `console.table` to display them in a table format.

STREAMOBJECT

The `streamObject` function is used for streaming structured data from an AI model and is useful for real-time updates. Similar to `generateObject`, it takes a Zod schema